Process Rewards for Mathematical Reasoning: GRPO Baselines and Failure Analysis

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Abstract

We compare outcome-only GRPO with process reward mixing for Qwen2.5-Math-1.5B. At about 6M generated tokens, outcome-only training reaches 54.00% on MATH500 and 67.02% on GSM8K; adding a frozen 7B PRM gives 53.80% and 64.97%. Neither difference is conclusive. Saved signals show PRM errors and normalization that gives low-scored wrong responses positive advantages. A bounded, correct-answer-only correction improves seed-17 answer accuracy. Seed 42 has smaller point gains, with both confidence intervals including zero. A matched-scorer control helps separate scorer adaptation from advantage construction.

1 Research question and motivation

Final-answer verification is cheap, but cannot distinguish valid reasoning from a lucky answer. Process supervision could help, yet an imperfect score can produce a misleading update. We ask whether PRM scores remain useful after GRPO transforms them into advantages, and whether the resulting policy solves more problems correctly.

Three findings guide the study. First, direct PRM mixing shows no clear accuracy benefit. Second, group normalization gives 577 wrong training responses positive advantages, including responses with low absolute PRM scores. Third, bounding the process correction gives promising seed-17 results, but seed 42 remains inconclusive. We also report the PRM adaptation pilot from our original proposal.

2 Related work and scope

GRPO learns from groups of responses without a learned critic (Shao et al., 2024); LoRA reduces the trainable parameter count (Hu et al., 2021). Human process supervision (Lightman et al., 2023) and Qwen's PRM study (Zhang et al., 2025) motivate step feedback, but also leave label reliability and distribution shift as concerns. ProcessBench tests first-error identification rather than only final-answer selection (Zheng et al., 2024). PURE assigns min-form step credit (Cheng et al., 2025); simply taking a minimum score and applying response-level GRPO, as in our B baseline, does not reproduce PURE.

The closest advantage methods are PAPO, which separately normalizes process signals within correct responses (Tan et al., 2026), and PROGRS, whose outcome-conditioned centering subtracts the incorrect-subset mean from wrong responses and leaves correct-response scores unchanged (Rezaei et al., 2026, their Eq. (2)). Our correction instead centers raw PRM scores only within correct responses and bounds its magnitude without a process standard deviation. PRPO also aligns process and outcome signals, at finer granularity (Ding et al., 2026). Our contribution is an empirical comparison and a bounded variant of these ideas. Dr. GRPO analyzes length bias in response-normalized losses (Liu et al., 2025), which we retain.

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3 Rewards and policy updates

3.1 Outcome verification and process scoring

For a problem q, the policy samples $G = 8$ chain-of-thought responses. Let $v_i = 1$ when response i ends normally and its answer is parseable under the fixed protocol. The binary outcome reward y_i is one if its final boxed answer matches the reference. Truncated or unparseable responses receive zero. For eligible responses, we split steps at canonical double line breaks, append Qwen's step marker, and read the positive-class probabilities:

$$\widetilde{m}_i = \begin{cases} \min_{1 \le k \le K_i} p_\phi(\text{valid} \mid q, s_{i,\le k}), & v_i = 1, \\ 0, & v_i = 0. \end{cases} \qquad R_i^A = y_i, \quad R_i^B = y_i + 0.2\widetilde{m}_i. \tag{1}$$

The zero for an ineligible response is a program placeholder, not a PRM judgment. B uses the unadapted Qwen2.5-Math-PRM-7B. C-mix uses the same reward formula with the adapted C scorer described in Section 6. Each scorer stays frozen during policy training.

3.2 GRPO objective

For each group, we compute one detached advantage per response:

$$\bar{R} = \frac{1}{G}\sum_j R_j, \qquad s_R = \sqrt{\frac{1}{G-1}\sum_j (R_j - \bar{R})^2}, \qquad \widehat{A}_i = \frac{R_i - \bar{R}}{s_R + \delta}, \quad \delta = 10^{-4}. \tag{2}$$

Let $h_{i,t} = (q, o_{i,<t})$, $\rho_{i,t} = \pi_\theta(o_{i,t} \mid h_{i,t}) / \pi_{\text{old}}(o_{i,t} \mid h_{i,t})$, and $u_{i,t} = \pi_{\text{ref}}(o_{i,t} \mid h_{i,t}) / \pi_\theta(o_{i,t} \mid h_{i,t})$. The old-policy denominator is detached, and the reference policy is fixed. With T_i active completion tokens, the selected `grpo` loss is

$$\begin{aligned} \mathcal{L} &= -\frac{1}{G}\sum_i \frac{1}{T_i}\sum_t \left[\ell_{i,t} - \beta\big(u_{i,t} - \log u_{i,t} - 1\big)\right], \\ \ell_{i,t} &= \min\Big\{\rho_{i,t}\,\text{sg}(\widehat{A}_i), \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\,\text{sg}(\widehat{A}_i)\Big\}. \end{aligned} \tag{3}$$

where $\epsilon = 0.2$, $\beta = 0.04$, and sg stops gradients (Shao et al., 2024; Hugging Face, n.d.). The reward need not be differentiable: a positive advantage increases the sampled tokens' log probabilities, while a negative one decreases them. Only policy LoRA parameters are updated. We use one optimizer iteration per rollout batch and no second advantage normalization.

3.3 Bounded process correction

For C-bounded, we first compute outcome advantages and then add a correction. Let $I_+ = \{j : y_j = 1\}$ and let $\bar{m}_+$ be the mean score over I_+. Define

$$\widehat{A}_i^{\text{CB}} = \frac{y_i - \bar{y}}{s_y + \delta} + c_i, \qquad c_i = 0.2\, y_i(\widetilde{m}_i - \bar{m}_+). \tag{4}$$

Set $c_i = 0$ when I_+ is empty, with no normalization after addition. Since scores lie in $[0, 1]$, $|c_i| \le 0.2$ and $\sum_i c_i = 0$. Wrong answers cannot receive positive reward advantages; all-wrong groups have zero reward advantages, although KL may still contribute a gradient. The correction therefore stays small even in homogeneous-outcome groups.

3.4 Common training settings

We use the base Qwen2.5-Math-1.5B (Yang et al., 2024), rank-16 LoRA (alpha 32, dropout 0), AdamW, constant learning rate 5×10^{-6}, and gradient clipping at 1. Each update uses four prompts and eight responses per prompt (microbatch 1, accumulation 32). Sampling uses temperature 1, top-p 1, and no top-k restriction. Prompt and completion limits are each 2,048 tokens. The implementation uses TRL 0.23.1, vLLM 0.10.2, and one A100 80 GB. Sampling importance correction is disabled.

4 Experiments and accuracy

We train on 6,993 MATH problems, hold out 500 for validation, and test on MATH500 (500) and GSM8K (1,319) (Hendrycks et al., 2021; Cobbe et al., 2021). Evaluation uses one greedy response, a 2,048-token cap, and the fixed verifier. GSM8K is evaluation-only. All intervals use 10,000 paired question-bootstrap samples (seed 285), measuring question-level rather than training-seed uncertainty.

4.1 Direct PRM mixing: historical V0 comparison

These runs use corrected percentage/unit verification and CRLF handling for PRM inputs; earlier buggy runs are excluded. A/B share initialization and 1,120 question groups, stopping after a complete update reaches 6M tokens. A uses 6,022,435 tokens and 284 updates; B uses 6,008,457 and 280. A has 16 extra groups: generation budgets are approximately matched, updates and total compute are not.

Table 1: V0, seed 17: outcome-only A versus unadapted-PRM B. Accuracy is in percent; differences and intervals are percentage points (pp).

Test	A	B	B−A	Paired 95% interval
MATH500	54.00	53.80	−0.20	[−4.00, +3.60]
GSM8K	67.02	64.97	−2.05	[−4.25, +0.08]

B does not show a clear benefit in this comparison. Checks regraded all 3,638 predictions. For 7,304 scored B responses, step spans and text hashes match PRM inputs; replayed advantages match loss inputs without duplicate normalization.

4.2 Adapted scorer and advantage construction: V1 comparison

V1 fixes 288 updates and question order within each seed. C-mix and C-bounded share the adapted scorer, isolating advantage construction. The initial model below uses V1 evaluation; it is not a matched V0 baseline.

Table 2: V1 accuracy (%). Tokens include failures; wrong-positive counts refer to training advantages. C-mix has only seed 17.

Seed	Policy	Tokens (M)	MATH500	GSM8K	Wrong-positive
—	Initial	—	49.00	56.86	—
17	A	6.155	53.00	66.87	0
17	C-mix	6.114	54.00	67.70	526
17	C-bounded	6.167	57.80	69.37	0
42	A	6.309	52.80	67.48	0
42	C-bounded	6.229	54.80	67.93	0

Table 3: V1 paired changes (pp, 95% intervals). Seed 42 is inconclusive; seeds are not pooled as independent test questions.

Seed	Comparison	MATH500	GSM8K
17	C-mix − A	$+1.00\ [-2.20, 4.20]$	$+0.83\ [-0.91, 2.58]$
17	C-bounded − A	$+4.80\ [1.40, 8.40]$	$+2.50\ [0.45, 4.55]$
17	C-bounded − C-mix	$+3.80\ [0.40, 7.40]$	$+1.67\ [-0.30, 3.64]$
42	C-bounded − A	$+2.00\ [-1.00, 5.00]$	$+0.45\ [-1.52, 2.43]$

Why there are two A results. V0 and V1 are independent runs with different stopping rules. V1 also enables prefix caching, chunked prefill, and asynchronous scheduling, with different vLLM memory allocation. Seed-17 follow-up restart histories differ between arms; seed 42 has matched restart boundaries. The two A scores (54.00%/53.00%) therefore do not isolate seed noise or one configuration change. We compare methods within protocols.

5 What goes wrong with the process signal?

5.1 A small reward weight need not produce a small advantage

When a group is all-correct or all-wrong, its outcome reward is constant. Substituting Eq. (1) into Eq. (2) gives

$$\widehat{A}_i^B = \frac{\widetilde{m}_i - \overline{\widetilde{m}}}{s_{\widetilde{m}} + \delta/0.2} = \frac{\widetilde{m}_i - \overline{\widetilde{m}}}{s_{\widetilde{m}} + 5 \times 10^{-4}}. \tag{5}$$

The 0.2 factor mostly cancels. A has zero reward advantages in such groups; B still ranks responses by the PRM. In V0 B, 242 of 1,120 groups (21.61%) have homogeneous outcomes. They contribute 19.92% of the absolute advantage sum, which is not a gradient-norm share. Of 577 wrong responses with positive advantages, 571 come from all-wrong groups and six from mixed groups.

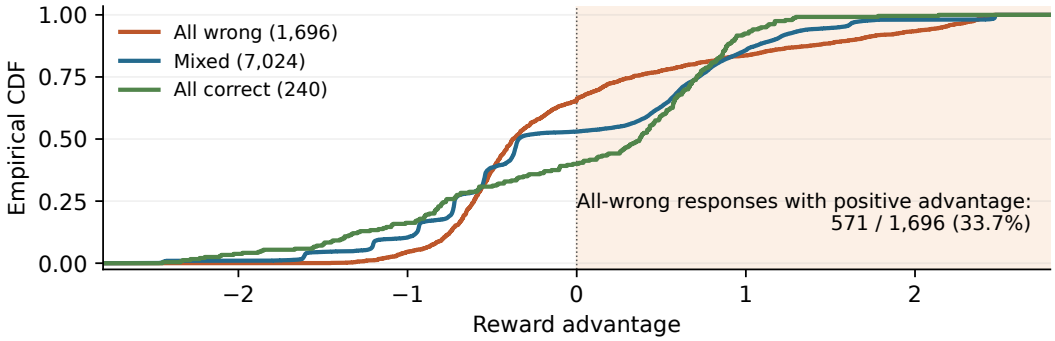


Figure 1: Actual V0 B advantages by group type, taken from the values passed into the policy loss. All-wrong groups alone give 571 of 1,696 responses (33.7%) positive advantages, with magnitudes comparable to the mixed groups. All curves use recorded training advantages.

5.2 Score errors versus misleading updates

Positive advantages for wrong final answers are not inherently harmful: a failed response can contain useful reasoning. We therefore inspect concrete, checkable errors rather than interpreting the counts alone. One saved response reports $4 \cdot 6 + 8 \cdot 3 - 28/2 = 36$ and claims its code returns 36; direct evaluation gives 34. Its PRM minimum is 0.9797, but its actual advantage is -1.072. Here the PRM misses an error, yet the update still penalizes the response.

Another response to a nine-vertex coloring problem uses node index 9, outside the valid range 0–8, and claims inconsistent counts of 132 and 12. Exhaustive enumeration of the original graph gives 54. Its minimum PRM score is only 0.0538, yet it receives advantage $+1.070$ in an all-wrong group. This shows how relative scaling can reward a clearly defective response without a high absolute score.

5.3 Formatting is a separate source of error

On GSM8K in V0, B has fewer terminated-but-unparseable outputs than A (53 versus 89), but more parsed wrong answers (368 versus 314) and truncations (41 versus 32). Missing answers alone do not explain the lower B score. Average step counts are similar (11.917/11.899), and PRM scores do not rise monotonically.

A separate rule-based MATH500 audit finds leading placeholder boxes in 8.2% of initial-model outputs, 30.0% of V1 A, and 63.2% of C-bounded (seed 17). A leading placeholder is an empty/template box within the first 20 non-whitespace characters, followed by at least 100 characters of body text. This measures output format, not when an answer was chosen. In 200 selected correct C-bounded outputs, deleting the placeholder slightly *increases* the mean minimum PRM score: original minus edited is -0.00445 (paired 95% interval $[-0.00890, -0.00009]$). That check does not support the simple explanation that the PRM rewards adding the placeholder.

6 PRM adaptation from the original proposal

The pilot uses 512 questions with four responses each and 2,048 shared step-label anchors, balanced positive/negative. C uses successful and failed responses; D uses failures alone. Both start from the same frozen 7B PRM and train rank-8 LoRA (alpha 16) for 512 matched updates at learning rate 2×10^{-5}.

Following LCA's pooling (Jia et al., 2026), define $e_{ik} = 1 - p_{ik}$ and $E_i = \sum_k w_{ik} e_{ik}$, with detached $w_{ik} = \mathrm{softmax}_k(e_i/0.2)$. C combines $-\log(1 - E_i)$ on successes, $-\log E_i$ on failures, and step-anchor cross-entropy; D removes only the success term. C/D are reward models, distinct from the policy variants C-mix/C-bounded. This pilot is not a full LCA reproduction.

Table 4: ProcessBench diagnostic: 64 valid and 128 invalid solutions. We show counts: acceptance requires all scores ≥ 0.5; an error hit requires the first score below 0.5 to match the label. H is the harmonic mean of the corresponding rates.

Scorer	Valid mean min	Valid accepted	First-error hits	H
Unadapted B	0.879	58/64	87/128	77.68%
C	0.842	56/64	94/128	79.85%
D	0.155	9/64	45/128	20.09%

D sharply reduces acceptance of valid derivations. C largely preserves it, although its difference from B is only two solutions. Adapting the scorer alone still leaves wrong-positive advantages under direct mixing (Table 2). D was not used for full policy RL.

7 Costs, limitations, and remaining experiments

Resource accounting. V0 takes 2.70 hours for A and 2.22 for B; peak allocated memory is 31.60 and 44.88 GiB. B scores 5.28M PRM input tokens (7.72M padded) in 593 seconds. In V1 seed 42, full A/C-bounded runs take 2.28/2.50 hours; C-bounded scores 5.59M input tokens in 749 seconds. For V1 seed 17, C-mix/C-bounded score 5.53/5.54M tokens in 1,681/2,071 seconds. Seed-17 continuation timers exclude earlier segments, so we do not present them as full-run times. PRM scoring times are included in training time; load varies on the shared GPU.

Limits of the evidence. Two training seeds do not establish reliable gains, and question-bootstrap intervals do not capture seed variation. A difference whose interval includes zero is unresolved, not evidence of equivalence. C-bounded changes gating, scale, and normalization together and discards process feedback on failed answers. Targeted cases do not estimate failure prevalence, and accuracy does not establish process validity. PRM exposure and validation overlap, response-length normalization, asymmetric seed-17 restarts, and disabled sampling correction remain limitations. Test outputs have been inspected; further design must use validation only.

Plan for the final report. By November 16, freeze evaluation code and complete a blinded review of 100 stratified derivations, reporting reviewer agreement and sampling weights. By November 23, run two controls *from the initial model*: shuffled within-correct PRM rankings and centered random corrections with matched magnitude. Each uses seed 17, V1, the same question order, and 288 updates (about 6M tokens), compared with the existing C-bounded run. Allow roughly 6–14 shared-GPU hours for both, plus evaluation; 48-update continuations will serve only as debugging checks. By November 30, finish analysis and, if resources allow, repeat the more informative control with seed 42. Dates are tentative.

Full training runs test the learning procedure, but do not reduce evaluation uncertainty by themselves. With 10% paired disagreement, a two-sided 5% test at 80% power needs about a 4.0-point difference on 500 questions (about 2.4 on GSM8K). Smaller effects may remain unresolved. Better accuracy and reviewed process validity than matched controls would support useful PRM feedback. Similar scores would leave the ranking contribution unresolved. Code and aggregate evidence are available online.[1]

[1] `https://github.com/mafeifei798/ece285-math-rl`

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